

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built containerized backend services and browser-automation workflows for MCA filings, adding retries, state recovery, CI/CD, and Google Cloud Run deployment paths.
- Optimized an AI-assisted company-registration pipeline and operational workflow around public registry lookups, generated drafts, and filing-state checks.
- Improved release velocity by **70%** by wiring Dockerized services to GitHub Actions and deployable cloud environments.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built medical-image segmentation tooling for **95 brain subparts** with reproducible preprocessing, inference, masks, and visualization outputs.
- Reduced segmentation runtime by **87%** while debugging correctness issues across large-volume data, generated masks, and 3D surface workflows.
- Contributed patches across mature open-source workflows while preserving compatibility with existing user paths.

Open Source

[Contributions](#) 

- Contributed patches across Invesalio, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working inside maintainer review processes for reliability, security, and integration quality.

Projects

Seaweed Contest Platform | *Go, PostgreSQL, Redis, Judge0, AWS, Docker, Kubernetes*

[GitHub](#) 

- Built a backend-heavy contest platform with async judging, score calculation, live rankings, runtime state, and deployment paths across Docker/Kubernetes on AWS.
- Modeled durable users/submissions/results in PostgreSQL and used Redis for low-latency contest state, keeping evaluation workflows responsive under concurrent contest load.

Murdoc | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#) 

- Built a self-hosted **AI security gateway** for LLM, HTTP-tool, and MCP traffic with route profiles, request scoring, runtime policy checks, and allow/block decisions before downstream execution.
- Added RBAC runtime settings, PII redaction, audit logs, attack-lab validation, OpenTelemetry traces, Prometheus metrics, and health checks for policy and failure visibility.

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus*

[GitHub](#) 

- Built Go/Echo backend services for a content distribution platform using browser automation, REST APIs, RabbitMQ workers, Redis autosave, and PostgreSQL persistence.
- Added retries, DLQs, circuit-breaker isolation, health checks, and Prometheus metrics so unreliable external browser/API workflows stayed recoverable across distributed workers.

Technical Skills

Languages: Go, Python, TypeScript, C++

Frameworks: FastAPI, Echo, Gin, Node.js

Tools: Kubernetes, Docker, PostgreSQL, Redis, OpenTelemetry, Prometheus, MCP, GitHub Actions, AWS, GCP